

Deep Dive — Pipelines, Multimodality, and Honest Evaluation

Part 05 — 🌈 Generative AI · 05-genai/notes/02-multimodal-and-practice.md

Note 01 covered the diffusion mechanism. This note covers what you actually build with it: the end-to-end text-to-image pipeline, video and audio generation, vision-language models, how to evaluate any of this (badly), and the failure modes and legal exposure you're signing up for.

1. Text-to-image, end to end

```
prompt: "a corgi astronaut, oil painting, dramatic lighting"
  |
  v
[text encoder] (CLIP text tower, or a full LLM/T5 encoder in newer systems)
  | a sequence of token embeddings, not one vector
  v
[diffusion model: U-Net or DiT] <-- negative-prompt embedding also feeds in (CFG)
  | cross-attention: every latent patch attends to every text token
  | repeated for N sampler steps (the scheduler picks step sizes / noise schedule)
  v
latent z' (denoised) --[VAE decoder]--> pixels
```

- **Text encoder.** Original Stable Diffusion used CLIP's text tower — pretrained via contrastive text/image alignment (Module 08 covers CLIP in the retrieval context), so its representations already carry visual-semantic structure. Newer systems (SD3, DALL-E 3, Imagen) add a full T5/LLM encoder in parallel because CLIP's 77-token limit and bag-of-concepts style struggles with compositional prompts — "a red cube on a blue sphere, left of a green cylinder" routinely gets attributes bound to the wrong object.
- **Cross-attention** is where text meaning enters the image: text tokens as keys/values, image-latent patches as queries. Same mechanism as Module 03, different job.
- **The scheduler** governs the noise schedule and step-update rule (DDIM, DPM-Solver++, Euler...) — swappable independently of the model weights, which is why one checkpoint behaves differently under different schedulers in ComfyUI/A1111.
- **Negative prompts** are not a separate mechanism — they're CFG (note 01 §2.4) where the "unconditional" branch is replaced by a second *conditional* branch on the negative text ("blurry, extra fingers, watermark"), and guidance pushes away from that instead of away from nothing. That's why negative prompts suppress specific defects so effectively: you're doing guidance twice, toward the wanted thing and away from a named unwanted one.
- **Seeds.** Initial noise x_t comes from a seeded RNG. With a deterministic sampler (DDIM), same seed + prompt + settings = same image — the basis of "seed hunting" as a real workflow.
- **img2img** starts from a real image partially noised to some intermediate x_t (via note 01's closed-form $q(x_t|x_0)$) instead of pure noise, then runs the reverse process from there. Lower starting t stays closer to the original; higher t gives more creative freedom.

- **Inpainting** is img2img where, at every reverse step, the *unmasked* pixels are reset to their correctly noised original values — the model can only "invent" inside the mask while staying consistent outside it.
- **ControlNet** (Zhang et al., 2023) adds an auxiliary network, trained alongside a frozen base model, that injects a structural signal (edge map, depth map, pose skeleton) into intermediate activations at every step. This decouples *what* (text controls content) from *where/how* (the control signal controls composition) — the single biggest unlock for production pipelines, because text alone is a poor interface for precise spatial control.

2. Video: same idea, harder substrate

Diffusing each frame independently fails immediately — independently sampled frames have no reason to agree frame-to-frame, so you get flicker, morphing objects, swimming backgrounds. Fixes make **time** a modeled dimension:

- **Temporal attention/convolution** layers interleaved with spatial ones, so the model looks across frames while denoising and keeps frame t and $t+1$ in agreement.
- **Spatiotemporal transformers** (Sora's approach): tokenize the whole clip as spacetime patches — a small cube of (height, width, time) rather than just (height, width) — and let self-attention handle spatial and temporal consistency uniformly. The DiT idea from note 01, with an extra axis.
- **Compute cost scales badly.** A 5-second, 24fps clip is ~ 120 frames — naively 120x an equivalent image's pixels, plus long-range temporal attention on top. This is why, as of 2026, video generation remains an order of magnitude more expensive per second than an equivalent-quality image, and why most production tools cap clip length and resolution well below text-to-image defaults.
- **Consistency failures are the diagnostic.** Temporal flicker, an object subtly changing identity mid-clip, physically implausible motion (objects interpenetrating, water not conserving volume) are video's version of six-fingered hands (§4): the model has learned appearance far better than physics.

3. Audio: three problems wearing one name

- **Text-to-speech.** Two families: **autoregressive** token models (predict discrete audio-codec tokens one at a time — Module 04's next-token idea over an audio vocabulary from a neural codec like EnCodec) and **diffusion** models that denoise a mel-spectrogram or waveform conditioned on text and a voice embedding. Autoregressive TTS tends to sound more naturally prosodic but is slower and can drift/repeat; diffusion TTS is more controllable and parallel but needs careful conditioning to avoid a flat cadence.
- **Music generation.** Same two families (MusicGen: autoregressive over discrete tokens; early Riffusion literally ran image diffusion on spectrogram images). Music adds a harder long-range structure problem — a coherent verse-chorus-verse arc over minutes is a much longer dependency than a spoken sentence, and it's where current systems are weakest.
- **Voice cloning.** Condition TTS on a short reference clip (seconds, in modern few-shot systems) to extract a speaker embedding, then generate new speech in that voice. The misuse surface is immediate: a few

seconds of anyone's voice — a voicemail, a video call — is enough for a usable clone. Direct line to [../13-ai-security/](#) (voice-based social engineering at scale).

4. Multimodal models: understanding vs. generating

Everything above *generates*. A separate capability is a model that *understands* images/audio well enough to reason about them — a different architecture problem. Conflating "can make pictures" with "can see" is a common, consequential mistake.

4.1 The LLaVA-style recipe

```
image --[frozen/lightly-tuned vision encoder, e.g. CLIP ViT]--> patch embeddings
                                     |
                                     [projection: linear or small MLP]
                                     |
                                     v
text tokens --[LLM token embeddings]--> concatenated sequence --[LLM, architecture unchanged]
```

The trick (LLaVA, Liu et al. 2023, and most 2023-era VLMs) is almost embarrassingly simple: run an image through a pretrained vision encoder (usually a CLIP ViT, already aligned to language via contrastive training) to get patch embeddings, learn a small **projection** into the LLM's own token-embedding space, and splice those projected "tokens" into the input sequence next to text tokens. The frozen (or lightly tuned) LLM's attention doesn't care what produced a token's embedding — it attends over the mixed sequence either way. What looks like a breakthrough ("the LLM can see!") is architecturally a small addition on top of work you already understand: a strong LLM (Modules 03-04) plus a strong aligned image encoder (CLIP).

4.2 Understanding and generation don't transfer

A model that accurately describes a photo (understanding, via an encoder feeding an LLM) has no built-in ability to *produce* a photo — that needs a generative decoder (diffusion, or an autoregressive image-token decoder) trained on a completely different objective. Historically these lived in separate products from the same lab (GPT-4V understood; DALL-E generated). Ask "can it see what I upload" and "can it make me an image" as two unrelated questions about any given system.

4.3 Where native multimodality is heading

The frontier direction (Gemini's natively multimodal training, GPT-4o's unified architecture, Chameleon) collapses the encoder-plus-projection pattern into a **single model trained end-to-end** on interleaved text/image/audio tokens from the start, rather than grafting vision onto a text-pretrained LLM. The bet: cross-modal structure learned *during* pretraining generalizes better than structure bolted on after. As of 2026 the gap is closing but not closed — bolted-on VLMs still lead on raw language reasoning (they inherit a mature text-only LLM); native multimodal models lead on tasks needing tight cross-modal binding (precise visual grounding, images that respect a subtle textual constraint).

5. Evaluation — and why every metric here is unsatisfying

Metric	Measures	Why it's not enough
FID	distance between real vs. generated feature distributions, via a pretrained Inception net	distributional similarity, not per-image quality or prompt fidelity; gameable; multiple incompatible "standard" implementations circulate
CLIP score	cosine similarity between image and prompt CLIP embeddings	rewards on-topic over correct — wrong object count or spatial relation can still score well if thematically close
Human preference / Elo	pairwise "which is better" votes, aggregated	expensive, slow, rater-pool bias; answers "which do people like," not "which is factually/physically correct"
Inception Score	class-confidence + diversity under an ImageNet classifier	mostly superseded by FID, still cited in older papers

None of these are satisfying because "good" for a generative model is multi-dimensional — realism, prompt fidelity, diversity, aesthetics, physical/factual correctness — and no scalar captures the mix a real product needs. This is the same unsolved problem [../15-ai-evals/](#) wrestles with for text: LLM-as-judge, human preference, and benchmark suites are all imperfect proxies for "is this actually good." Generative image/audio/video evaluation is arguably the harder version, because the space of "correct" outputs for a creative prompt is legitimately not single-valued.

The diagnostic that actually works: look at hands and text. Diffusion models learn statistical regularities from pixels; hands have high-degrees-of-freedom structure that's hard to learn robustly from 2D projections, and rendered text requires exact discrete symbol sequences from a continuous generative process with no notion of "spelling." Both remain reliable tells for "likely diffusion-generated" well into 2026, even as overall photorealism has become hard to distinguish from real photography otherwise.

6. Failure modes, provenance, misuse

- **Mode collapse (GANs).** The generator finds a small set of outputs that reliably fool the discriminator and stops exploring. Diagnostic: sample 100 seeds and look for near-duplicates. Diffusion is far more resistant by construction — no adversarial game to "win" by narrowing.
- **Hallucinated detail** is the image analogue of an LLM hallucination: confidently rendered detail that's locally plausible and globally wrong (garbled not-quite-text on a sign, a sixth finger, an architecturally impossible building). Same root cause as Module 04 §9: trained to produce *plausible* continuations of the noising process, not to verify correctness against ground truth.
- **Watermarking and provenance — C2PA.** The Coalition for Content Provenance and Authenticity standard embeds cryptographically signed metadata ("Content Credentials") recording how an image was created/edited. Invisible statistical watermarks (Google's SynthID is the most deployed) embed a signal in the pixel/token distribution that survives resize/compress/screenshot and is detectable without the original. **Neither is a complete solution:** C2PA metadata is trivially stripped by any tool that doesn't

preserve it (a plain screenshot does), and no deployed watermark is provably robust against a motivated adversary who wants it gone — both are friction layers, not cryptographic guarantees.

- **Deepfakes and misuse.** Voice cloning (§3) and face/video synthesis push impersonation and non-consensual synthetic media to near-zero marginal cost. Direct line to [../13-ai-security/](#) (synthetic-media fraud, cloned-voice social engineering) and [../12-ai-governance/](#) (disclosure obligations, deployer responsibility, the emerging — in some jurisdictions regulatory — expectation that AI-generated content be labeled).

7. Licensing and copyright — two separate questions

1. **Was the training data licensed?** Most large image models trained on web-scraped datasets (LAION-style) whose images weren't individually licensed for this use — the subject of active, unresolved litigation across jurisdictions (artists and stock-image companies vs. model providers) as of 2026. The legal theory (fair use / text-and-data-mining exceptions vs. infringement) hasn't settled uniformly.
2. **Who owns the output, and is it infringing?** The US Copyright Office's current position: purely AI-generated output without sufficient human creative authorship is **not copyrightable** — you can't register a bare prompt-and-click image. Separately, an output can *infringe* an existing work (a recognizable protected character, a style tightly tied to one living artist, memorized training data) regardless of who or what generated it — generativeness isn't an infringement exemption on the output side.

Do not assume "the model made it" resolves either question. Check the license terms of the specific model/API (many commercial APIs contractually grant you usage rights over outputs regardless of the training-data question — that's the provider absorbing risk, not proof the underlying question is settled), and treat outputs resembling a specific living artist's style or a copyrighted character as real legal exposure. Ties directly to [../14-ai-compliance/](#) for building an audit trail around model/data provenance and usage terms.

8. Forward and backward links

Idea here	Where it connects
CLIP-style contrastive alignment	Module 08 — the same text-image alignment powers multimodal RAG
Evaluation being unsatisfying	Module 15 — the text-generation version of the identical problem
Deepfakes, voice cloning misuse	Module 13 — AI security, social-engineering threat models
Disclosure, provenance obligations	Module 12 — AI governance
Training-data licensing, output copyright	Module 14 — AI compliance, audit-ready evidence
ControlNet-style external conditioning	Module 06/07 — agents/tools calling generation APIs with structured control inputs
Diffusion mechanism (forward/reverse/CFG/DiT)	notes/01-core-concepts.md [05-genai/notes/01-core-concepts.md]